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Attentional biases to emotion in impulsive and instrumental violent offenders: an event-related potential study

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Background: Interpersonal violent behavior is often induced by extreme emotional states but the effects of emotional stimuli on this behavior remain poorly understood. *Aim:* We compared the effects of emotional stimuli on behavior in individuals with a history of impulsive or instrumental violence to test for either general problems in processing of emotional stimuli or problems in disengagement from this type of information. *Methods:* Behavioral and electrophysiological data were collected using emotional stimuli preceding a Posner Cuing Task. *Results:* Impulsive violent individuals showed more attentional resource allocation to angry faces whereas instrumental violent individuals attended less to sad faces. Electrophysiological measures showed there were differences in attentional allocation leading to differences when disengaging from invalid emotional cues, rather than a pattern indicating differences in early processing in extrastriate cortex. *Conclusions:* Different emotion–attention problems are seen in impulsive and instrumental violent offenders due to abnormalities in disengaging attention from specific types of emotional content.

Keywords: impulsive violent behavior; instrumental violent behavior; attentional bias; emotional Posner's Cuing Task; ERPs

Introduction

Violence is pervasive in society. This behavior may be subdivided into impulsive and instrumental violence, with the former elicited in response to provocation and frustration and the latter related to inflicting serious injury on a person for goal-directed purposes (Cornell et al., 1996). Importantly, these two types of violent behavior can be reliably distinguished (Cornell et al., 1996) and, as mentioned by Swogger, Walsh, Houston, Cashman-Brown, and Conner (2010), such a distinction can be useful in determining the likely course of treatment (Eaves, Douglas, Webster, Ogloff, & Hart, 2000). These subgroups therefore

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offer possible insight into contributory factors and mechanisms related to violent behavior.

Impulsive violent offenders typically show aggressive responses to provocation and threat. They have either a reduced cortical volume or show decreased activation of the frontal areas that are involved in inhibitory control. Additionally, they show impaired regulation of negative emotions such as anger and fear (Raine et al., 1998). One mechanism suggested to account for this is the social response reversal system. This system may be activated by aversive social cues such as angry faces and is associated with the modulation of reactive aggression (Blair, 2004). The social response reversal theory states that individuals, low in a hierarchy, should be more susceptible to anger and aggression from higher ranked peers (Blair & Cipolotti, 2000). High-ranked individuals would be expected to react with aggression in response to anger from someone with a lower status. This suggests a role for social hierarchy in aggression, but also predicts that impulsive violent individuals who perceive themselves as high-ranked will be more sensitive to angry facial displays and react aggressively to maintain their status.

Instrumental violence is thought to be a characteristic of psychopathic adult offenders (Meloy, 1992; Meloy & Gothard, 1995; Serin, 1991; Williamson, Hare, & Wong, 1987; Woodworth & Porter, 2002). For example, Woodworth and Porter (2002) showed that instrumental homicides tend to be committed more frequently by psychopathic offenders (Woodworth & Porter, 2002), such that the vast majority of their homicides are of this type. Instrumental aggression is also linked to impairments in empathy and moral socialization (Blair, 1995, 2001; Blair, Morris, Frith, Perrett, & Dolan, 1999; Roth & Strüber, 2009). For this type of behavior, rather than the interaction of social hierarchy with emotional state, the fear dysfunction, and violence inhibition mechanism (VIM) hypotheses may be informative. The VIM was suggested by Blair (1995) who argued that normally, distress cues may lead to withdrawal responses and the termination of aggressive behavior in order to avoid negative feedback cues from a victim. Instrumental violent offenders have been suggested to have impairments in the processing of fearful and sad expressions which may be one component of such distress cues. A decreased sensitivity to these expressions in others would be a characteristic of problems with the VIM, with weaker associations between actions and punishment or negative feedback.

Thus, previous work leads to the idea that impulsive and instrumentally violent individuals may have difficulties in emotional processing and that these difficulties differ systematically between the two groups. One possibility is that such deficits may be a consequence of issues with processing information at the attentional focus. Impulsive violent individuals may be more engaged by angry information and instrumental violent individuals underengaged by fearful or sad information. However, only a relatively small number of studies have investigated attentional control in impulsive violent offenders (Chen, Muggleton, Juan, Tzeng, & Hung, 2008; Chen, Tien, Juan, Tzeng, & Hung,

2005; Hoaken, Allaby, & Earle, 2007). Additionally, no study has directly compared impulsive and instrumental violent offenders on attentional control tasks and investigated their underlying neural correlates. The combination of behavioral testing with electrophysiological recording, such as event related potentials (ERPs, electrophysiological components that may be evoked by stimuli or provide measures of neural processing), offers the potential for insight into neural mechanisms, as indexed by differences in specific ERP components in these individuals.

The characteristics of the general behavior of these two groups are different in many ways. These are not limited to the manner and reasons thought to underlie their violent behavior and may include more general characteristics of the populations (Stanford, Houston, & Baldridge, 2008). Few studies have investigated the interaction between emotion and attention in general, and their manifestation in the two different types of violent offenders specifically (c.f. Newman, Curtin, Bertsch, & Baskin-Sommers, 2010).

We used an emotional Posner's Cuing Task to investigate the effects of emotion on attentional processing in impulsive and instrumentally violent individuals to evaluate whether they showed general or specific attentional abnormalities. This task has been used to investigate attention in anxiety disorders (e.g. Fox, Russo, Bowles, & Dutton, 2001; Li, Li, & Luo, 2005; Van Honk et al., 2003) and trait anger has been found to bias performance for threat-related materials in such cognitive tasks (Cohen, Eckhardt, & Schagat, 1998; Eckhardt & Cohen, 1997; Van Honk, Tuiten, de Haan, van den Hout, & Stam, 2001; Van Honk et al., 2001). It was found that individuals with high trait anger showed an attentional bias for angry facial expressions (Van Honk, Tuiten, de Haan, et al., 2001). Such an attentional bias could be a result of either enhanced orienting/engagement towards stimuli or delayed disengagement from threat.

Participants were a forensic sample of violent offenders, allowing exploration of the information-processing correlates of anger, fear and sadness in impulsive and instrumental violent individuals in comparison to each other and to controls. There are two main possibilities to explain any differences that may be seen as a consequence of emotional stimuli. First, there may be differences in the processing of these stimuli. Impulsive violent offenders may over-process aggressively themed materials and angry facial expressions and have more difficulty disengaging from anger-related stimuli. Instrumental violent offenders may also fixate on anger-related stimuli but would be expected to be more able to control/plan their reaction. These offenders would be predicted to under-process sad and fearful faces because sadness and fear are less relevant to their own emotional experience (Clark, Fiske, & University, 1982). Second, any differences in patterns of performance could reflect differences in disengagement from emotional stimuli, based on the presupposition that latency to disengage from a cue is a function of the depth of processing or strength of engagement of that cue (Derryberry & Reed, 2002; Fox, Russo, & Dutton,

2002). Problems with disengagement would also be consistent with the view that impulsive violent offenders may be less able to use strategic attentional control once signals of interpersonal threat have been engaged. To investigate these possibilities, ERPs were used as indicators of the time-course of attentional deployment, similar to use of this technique in selective attention tasks (e.g. Hillyard & Anllo-Vento, 1998). Such studies of spatial selective attention have identified specific indices of attention allocation in a number of ERP components such as the P100, N100, N2pc, and P300.

The P100 and N100 components are dissociable in that the occipital P100 reflects correctly directed spatial attention and the N100 reflects discrimination of attended stimuli (Mangun, 1995; Mangun & Buck, 1998). Thus, according to the possible differences in offenders, the following would be expected. First, increased allocation of attention to negative facial expressions in impulsive violent offenders would be predicted to result in attention-related increases in the amplitude as well as faster onset latencies of these early ERP components. In contrast, if the instrumental violent offenders pay less attention to emotional expressions, particularly sad faces, slower P100 and N100 latencies, and lower P100 and N100 amplitudes to sad stimuli would be predicted for these participants.

There is increasing evidence that the parietal N2pc component is involved in attentional selectivity for affective expressions. This component is usually seen between 170 and 300 ms after stimulus onset contralateral to the side of an attended visual stimulus (Eimer & Kiss, 2007; Holmes, Bradley, Kragh Nielsen, & Mogg, 2009). Increased allocation of attention to angry facial expressions by impulsive violent offenders would be expected to result in higher amplitude of the N2pc. The instrumental violent offenders might pay less attention to emotional expressions, particularly sad faces, so an effect in the opposite direction may result.

The P200 component (the major positive deflection occurring 120–315 ms after a cue) was previously investigated by Carretié, Mercado, Tapia, and Hinojosa, 2001. They found an increased P200 amplitude and faster P200 latency in response to negatively vs. positively arousing pictures. Thus, the P200 evoked by negatively arousing face cues (especially anger), should they be processed differently, may be larger and faster for impulsive violent offenders than for normal controls. The P200 evoked by the negatively arousing face cues (especially sad faces) in the instrumental violent offenders, in contrast, may be smaller and slower than normal controls.

The P300 component is thought to reflect central neural system activity involved in attentional resource allocation (Jennings & Coles, 1991; Polich & Kok, 1995) with the P300 source observed mainly found to be the anterior cingulate cortex (ACC) (Bokura, Yamaguchi, & Kobayashi, 2001; Maeno, Kaneko, Iramina, Eto, & Ueno, 2003). Here, the amplitude of P300 was taken as being indicative of attentional allocation generated from ACC, an area linked to the regulation of aggression (Stadler et al., 2007), although it is important to

note that ERPs are both correlational and there is frequently ambiguity regarding the source of activity. The ACC can be anatomically subdivided into two parts. The gyrus (ACCg) has been shown to be involved in social/affective processing and is also linked to the amygdala and to the orbitofrontal cortex. The sulcus (ACCs) is explicitly involved in non-social/affective processing and is linked to motor areas (Behrens, Hunt, & Rushworth, 2009; Rushworth & Behrens, 2008). Irrespective of the ERP source information, should the impulsive violent offenders show differences from normal subjects in disengagement from angry stimuli, they would be expected to show a selective decrease in P300 in invalidly cued trials with angry stimuli, reflecting decreased resources required to disengage from the previously cued location. In other words, impulsive violent offenders would devote relatively less processing resources to disengaging attention from angry, but not sad, cues. Conversely, the instrumental violent offenders might show a higher P300 to the target in sad invalid trials, indicating less engagement by these faces in these individuals.

This can essentially be summarized as changes in later ERP components being more indicative of attentional differences related to emotional stimuli, whereas early components may reflect general differences that relate to the processing of such stimuli.

In summary, we aimed to investigate the effects of emotion on attentional processing in impulsive and instrumentally violent individuals in conjunction with electrophysiological recordings to evaluate whether they showed general or specific attentional abnormalities. In doing so, we aimed to test whether there were problems with the processing of emotional stimuli or problems with disengagement from such information.

Methods

Overview

Experiment 1 consisted of an Emotional Posner's Cuing Task presented to impulsive and instrumental violent offenders and to controls in order to investigate whether they differed in terms of task performance following presentation of emotional content. Thus, this experiment aimed to evaluate the behavioral consequences of the emotional content of the cuing stimuli. For Experiment 2, the Posner's Cuing Task was presented and electrophysiological activity recorded to evaluate the time course of spatial attention and resource allocation to the emotional cue-faces in both violent groups and in controls.

Participants

Experiment 1: Posner's Cuing Task (behavior)

Eight volunteer male impulsive violent offenders (mean age 31.4 ± 2.93 years), eight instrumental violent offenders (mean age 29.9 ± 3.39 years), and eight

matched controls took part in this task (age 30.3 ± 1.83 years). The standard version of the Raven's Progressive Matrices test (Raven, Raven, & Court, 1993) was also presented to subjects. This provides a measure of high-level analytical reasoning and fluid intelligence (e.g. Carpenter, Just, & Shell, 1990) and is independent of reading and writing skill (see, e.g. Hayashi, Kato, Igarashi, & Kashima, 2008). No significant differences were seen between the three groups on this test (mean score 17.0 ± 6.63 in the impulsive violent group, mean score 16.8 ± 5.21 in the instrumental violent group, and mean score 16.5 ± 5.55 in the control group, $F(2, 21) = .01$, $p = .99$). There were also no differences among the three groups in education duration (mean years of education 9.8 ± 1.39 in the impulsive violent group, 10.5 ± 1.60 in the instrumental violent group, and 10.9 ± 1.50 in the control group, $F(2, 21) = 1.14$, $p = .34$). Groups did not differ in IQ or socioeconomic status.

Experiment 2: Posner's Cuing Task (electrophysiology and behavior)

Ten volunteer male impulsive violent offenders (mean age 33.2 ± 5.25 years), ten instrumental violent offenders (mean age 30.2 ± 4.21 years), and ten matched controls participated in the ERP experiment (mean age 30.8 ± 2.04 years). Eight of the ten impulsive violent offenders and eight of the ten instrumental violent offenders had also taken part in Experiment 1. There were no differences among the three groups in the Raven's Progressive Matrices test (mean score 18.6 ± 6.46 in the impulsive violent group, 16.7 ± 7.55 in the instrumental violent group, and 16.0 ± 5.20 in the control group, $F(2, 27) = .25$, $p = .78$). There were also no differences for education duration (mean education years 9.6 ± 1.26 in the impulsive violent group, 10.2 ± 1.55 in the instrumental violent group, and 10.5 ± 1.58 in the control group, $F(2, 27) = .97$, $p = .39$). Groups did not differ in IQ or socioeconomic status.

For both Experiment 1 and Experiment 2, all participants were categorized as impulsive violent offenders, instrumental violent offenders or non-violent offenders based on their prison records, self-reports of previous criminal history and official records of the nature of their criminal behavior. Subjects were selected from those without any prior diagnosis of a psychiatric disorder or any history of drug abuse.

Impulsive violent offenders were defined as those with a history of impulsive violent offenses but no history of instrumental violent offenses. They had all committed homicides, and were repeat offenders with the homicides all occurring on separate occasions. All were serving sentences of fifteen years to life in prison. The classification of their behavior as impulsive was based on the fact that there was no evidence that they used violent behavior to achieve a specific desired goal, such as obtaining the victim's possessions or to increase status within a group hierarchy.

The instrumental violent offenders had committed at least one violent crime, which, unlike the impulsive violent individuals, occurred with a clear

purpose other than reacting to provocation or frustration. They had committed premeditated robbery-homicide, and had specifically planned to use knives, guns or sticks to result in either the death of another person or to inflict serious body injuries. The instrumental violent offenders had served more than 15 years' imprisonment or had been sentenced to life imprisonment.

In both groups of offenders, only individuals who had shown only the type of behavior appropriate for their classification were included. As such, for example, instrumentally violent individuals who had also been involved in impulsive violence were not included. Individuals for which there was ambiguity in their classification (as a consequence of the nature of their criminal record were not included).

The classification of both groups was consistent with Cornell et al. (1996) who defined instrumental violence as goal-driven, requiring planning and involving no provocation, in contrast with reactive aggression which occurred in the absence of planning or goals and typically involved dispute or interpersonal conflict. Table 1 shows the comparison of impulsive and instrumental violent offenders in terms offending behaviors.

Neither group of violent offenders included murderers who committed robbery or rape, nor any involved in domestic violence or sexual offences.

The control non-violent offenders had no convictions relating to violent offenses and no indication of being involved in any assault. They had also all committed offenses, but not of a violent type. Offenses they were convicted of included counterfeiting documents, counterfeiting securities, larcenies, fraud, and accepting bribes as a public servant.

While there is evidence supporting this type subtyping of violence (Barratt, Stanford, Dowdy, Liebman, & Kent, 1999; Haden, Scarpa, & Stanford, 2008), and there has been differentiation of subtypes for a range of populations (e.g. violent men; Stanford, Houston, Villemarette-Pittman, & Greve, 2003, college students; Haden et al., 2008; see Bo et al., 2013), there may be issues, particularly when there are common features shared by different groups (Bo et al., 2013). Within the context of the current study, it would be expected that, should there be insufficient separation of the groups investigated, or the differentiation not be biologically meaningful, the result would be no difference in the patterns of performance or electrophysiological data for the two violent offender groups.

This method of classifying offenders based on their criminal behavior, particularly for individuals who have committed multiple homicides, is likely to result in accurate classification. For example, Woodworth and Porter (2002) found that those who were involved in instrumental violence (classified as psychopaths by means of the Psychopath Checklist-Revised (Hare, 1991 in their study) rarely employed reactive aggression in homicides. This suggests that classification based on crimes of this type should be efficient in distinguishing instrumentally violent individuals from those who are impulsively violent; particularly when the crimes they were involved in had to all belong to exclusively one of the two types.

Table 1. The characteristics of the impulsive, instrumental violent, and control groups and the number who took part in each part of the experiment. The offenders were classified by official records and self-report of past criminal history (interview) and the offences are summarized briefly in the table. Importantly, the characteristics of the crimes for the groups dictated the participant classification as either instrumental or impulsive offenders (i.e. all of the instrumental offenders had carried out premeditated crimes whereas the impulsive violent offenders had committed 'spur of the moment' offences).

	Brief description of criminal activities	Number of subjects	
		Posner's Cuing Task: behavioral experiment	Posner's Cuing Task: ERP experiment
Impulsive violent group	(1) Committed offenses causing bodily harm	4	4
	(2) Committed homicide		
	(3) More than fifteen years imprisonment		
	(1) Committed homicide twice	4	6
	(2) Sentenced to life-time imprisonment		
Instrumental violent group	(1) Convicted of theft, robbery, and homicide	5	4
	(2) More than fifteen years imprisonment		
	(1) Committed homicide with a gun	2	4
	(2) More than fifteen years imprisonment		
	(1) Committed homicide for insurance benefits	1	2
	(2) Sentenced to life-time imprisonment		
Control group	Counterfeited documents	3	4
	Counterfeited documents and convicted of larceny	2	2
	Counterfeited securities	2	2
	Accepting bribes as a public servant	1	1
	Convicted of fraudulence	none	1

All participants were matched for age, education, and gender and were recruited from Taipei Prison in Taiwan. All had normal or corrected-to-normal vision, gave written informed consent before participating in the study, and both experiments were approved by the ethics committee of Taipei Prison, Taiwan.

Methods: Posner's Cuing Task (behavior)

The version of Posner's Cuing Task used employed images of faces ($2.5^\circ \times 2.5^\circ$) that could show negative (anger, fear, sad), positive (happy), or neutral expressions. These acted as cues for the target, a black circle of diameter $.3^\circ$. A trial went as follows: following a 500 ms fixation a face cue was presented for 250 ms in a $2.5 \times 5^\circ$ box 2° to the left or right of the fixation. This was blanked out for 50 ms and the target was then presented in the lower half of one of the two boxes until a key press response indicating the target location was made (or until 2000 ms had elapsed). The cue location was valid on 75% of trials and invalid for 25% of trials. 384 trials were presented in blocks of 96 trials and each cue emotion appeared equally frequently. Cue validity effects were calculated as mean RTs for invalid minus valid trials (i.e. disengagement cost). A 3 (Group: impulsive violent, instrumental violent, and control) $\times$ 2 (Cue validity: valid and invalid) $\times$ 4 (Cue valence: angry, fearful, sad, and neutral) mixed design ANOVA was used. Group was a between-subjects factor while cue validity and cue valence were within-subjects factors. The task is illustrated in Figure 1.

Methods: Posner's Cuing Task (electrophysiology and behavior)

The design of the task when electrophysiology recordings were made was the same as in Experiment 1 except that the faces used had only negative (angry or sad) or neutral expressions. 320 trials were presented in 80 trial blocks with 75% cue validity and equal proportions of each type of face and target location. EEG activity was recorded with Ag/AgCL electrodes mounted in an elastic cap (Electrocap International) using a 38-electrode arrangement following the International 10–20 system, referred to the left mastoid and with a ground electrode on the forehead. Vertical and horizontal electrooculograms were also recorded and electrode impedances were kept below 5 k Ω . Data were recorded with Neuroscan 4.2 software with a sampling rate of 500 Hz.

Analysis consisted of the following steps. (1) Epoching with cue onset set as the zero point and epochs running from -100 ms for 800 ms. (2) Filtering with a .05–30 Hz bandpass filter. (3) Baseline correction by the mean amplitude of the period 100 ms before target onset. (4) Artifact rejection to exclude trials with eye blinks (EOG amplitude greater than ± 50 μ V) or with movement artifacts (EEG greater than ± 100 μ V). (5) Averaging for each condition, subject, block and stimulus category for the F3, Fz, F4, C3, Cz, C4, P3, Pz, P4,

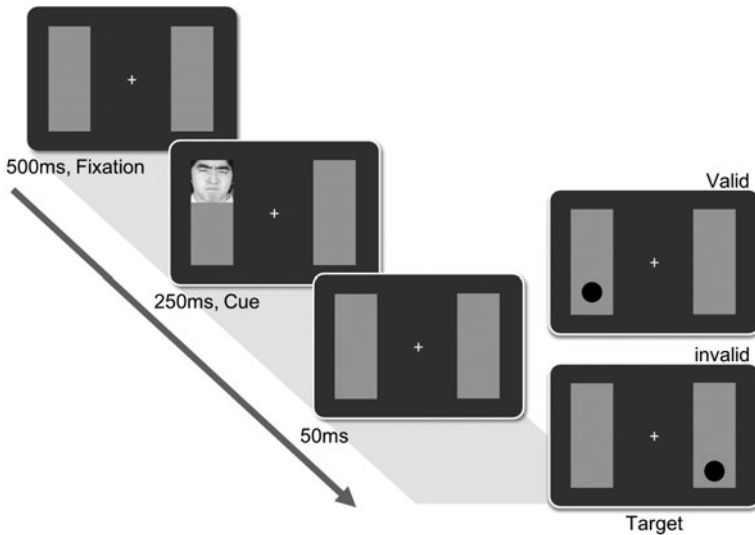


Figure 1. The Posner's Cueing Task used in the experiment. Following a fixation, faces were presented briefly (250 ms). These face cues were selected from negative (angry, fearful or sad), positive (happy), and neutral facial expressions. Following a 50 ms interval the target of a black circle was presented. Both cue and target were presented inside one of the two gray rectangular boxes which were displayed 2° to the left and the right of the central fixation point. The face cue correctly predicted the target location side in 75% of the experimental trials (valid cue trials). In the remaining 25% of the trials, the target appeared in the opposite location to the cue (invalid cue trials). The subject had to make a keypress to indicate the location of the target stimulus and response times were collected. Disengagement costs were calculated as the mean response times to the target for invalid trials minus mean response times for valid trials.

O3, Oz, O4, P7, and P8 electrode sites. (6) Grand averages over blocks (individual grand-averages) and subjects (across-subject grand-averages) were calculated. For these F3, Fz, and F4 were selected as the frontal area, C3, Cz, and C4 as the central area, P3, Pz, and P4 as the posterior area, and O3, Oz, and O4 as the occipital area. For the N2pc component, P7 and P8 were used, with the difference between the two representing the component.

Results

Experiment 1: behavioral task

Table 2 shows the reaction times for the angry, sad, fearful, happy, and neutral cues face for the impulsive violent, instrumental violent groups, and the matched controls on the Posner Cuing Task. Analysis of RTs used ANOVA with factors of subject group, cue validity, and cue valence. *Post hoc* analysis for both this experiment and Experiment 2 employed the method of least

Table 2. Posner's Cuing Task: mean correct response times and mean disengagement costs (in milliseconds) for the five types of face cue (angry, sad, fearful, happy, and neutral) in impulsive, instrumental violent, and control groups. Disengagement costs were computed as mean response times for invalid trials minus mean response times for valid trials. The disengagement score with angry face cues for the impulsive violent group was larger than that for the control group. The disengagement score with sad face cues for the instrumental violent group was smaller than that of the control group and the impulsive violent group.

Group	Face cue		
	Angry Mean RT (SD)	Happy Mean RT (SD)	Neutral Mean RT (SD)
<i>Impulsive violent group</i>			
Valid trials	289.2 (21.6)	290.4 (24.1)	292.9 (25.5)
Invalid trials	349.5 (32.8)	338.1 (36.7)	338.3 (34.7)
Disengagement cost	60.3 (20.7)	47.7 (22.1)	45.4 (20.6)
<i>Instrumental violent group</i>			
Valid trials	252.0 (28.5)	253.0 (29.9)	252.3 (28.0)
Invalid trials	302.5 (33.8)	304.3 (33.1)	308.4 (28.7)
Disengagement cost	50.5 (8.6)	51.3 (12.9)	56.1 (10.3)
<i>Control group</i>			
Valid trials	276.7 (24.1)	274.8 (23.0)	273.8 (23.3)
Invalid trials	321.3 (26.7)	321.1 (28.2)	323.1 (22.7)
Disengagement cost	44.6 (10.7)	46.3 (10.7)	49.3 (6.1)

significant differences along the lines of Saville (1990). Levene's test was used to test whether population variances differed significantly and Cohen's d was used as a measure of effects size. For the latter, values greater than .8 are typically interpreted as 'large' effects. A significant Group $\times$ Cue Validity $\times$ Cue Valence interaction effect was seen ($F(8,84) = 4.41, p < .001$). For valid trials, there was no significant group $\times$ Cue valence interaction. For invalid trials, there was a significant group $\times$ Cue valence interaction ($F(8,84) = 6.94, p = .005$). In invalid trials, the mean RT for angry face cues in the impulsive violent group was longer than matched controls ($t(14) = 2.823, p = .014$; Levene = .123, $p = .731$; Cohen's $d = .943$). For the impulsive violent group, the mean RT for angry faces in the invalid trials was significantly longer than those of fearful ($t(7) = 4.10, p = .01$), happy ($t(7) = 3.79, p = .01$), and neutral faces ($t(7) = 3.30, p = .01$).

There was a significant Group $\times$ disengagement score (angry, sad, fearful, happy, and neutral) interaction ($F(8,84) = 4.41, p < .001$) (the disengagement score was the mean RTs for invalid trials minus that for valid trials). The disengagement score with angry face cues for the impulsive violent group (60.3 ± 20.7 ms) was larger than for the control group (44.6 ± 10.7 ms), approaching significance ($t(10.502) = 1.913, p = .08$; Levene = 6.018, $p = .028$; Cohen's $d = .9628$). The disengagement score with sad face cues for the instrumental

violent group (33.9 ± 6.9 ms) was smaller than that of the control group (53.9 ± 12.7 ms) ($t(14) = 3.911$, $p = .002$; Levene = 3.571, $p = .08$; Cohen's $d = 1.9569$) and the impulsive violent group (54.7 ± 11.2 ms) ($t(14) = 4.52$, $p < .001$; Levene = 2.085, $p = .171$; Cohen's $d = 2.2362$). For the impulsive violent group, the disengagement score with angry face cues was larger than those with happy ($t(7) = 2.83$, $p = .03$) and neutral ($t(7) = 2.51$, $p = .05$) face cues. For the instrumental violent group, the disengagement score with sad face cues was smaller than those with angry ($t(7) = 4.69$, $p = .002$), fearful ($t(7) = 2.71$, $p = .03$), happy ($t(7) = 6.03$, $p = .001$), and neutral ($t(7) = 8.22$, $p = .001$) face cues.

Experiment 2: behavioral and ERP task

Behavioral data from the Posner Cuing Task showed an elevated mean RT for angry face cues in the impulsive violent group ($F(1,18) = 5.30$, $p = .03$). The disengagement score with angry face cues was larger than that with neutral face cues in the impulsive violent group ($t(5(9)) = 3.87$, $p = .004$). For invalid trials, the mean RT for sad face cues was lower than for neutral cues for the instrumental violent group but this difference did not reach significance ($t(5(9)) = -2.04$, $p = .07$).

Analysis was carried out separately on the cue-evoked P100 occipital component, N100 occipital component, and the P200 frontal component each using a 3 (Group: impulsive violent, instrumental violent and control) $\times$ 2 (Cue valence: angry and sad) mixed design ANOVA. The assumption of sphericity was not violated (Mauchly test for sphericity $p > .05$) so no correction was necessary. The analysis was performed separately for the cue-evoked early and late N2pc component on angry trials with the difference wave being the result of contralateral angry face cue minus the ipsilateral angry face condition in this case. The difference wave of the N2pc component was analyzed using a 3 factor (Group: impulsive violent, instrumental violent, and control) one-way ANOVA. The target evoked P300 component was analyzed for valid trials using a 3 (Group: impulsive violent, instrumental violent, and control) $\times$ 2 (Cue valence: angry and sad) $\times$ 3 (Electrode site: frontal, central, and posterior) mixed design ANOVA. In this case, the dependent variable was the mean amplitude of the P300 difference wave (i.e. angry valid minus neutral valid and sad valid minus neutral valid components). The same analysis was performed for the P300 component on invalid trials with the difference wave being the result of the angry invalid minus neutral invalid condition, and sad invalid minus neutral invalid condition in this case. For these analyses, the dependent variable was the mean component amplitude. The intervals for the various components were: P100, 150–190 ms after cue onset; N100, 200–240 ms; P200, 190–230 ms; early N2pc, 170–220 ms; late N2pc, 220–270 ms. To investigate N2pc differences, the mean amplitude of the difference waves was computed by subtracting ERPs at the electrode ipsilateral to the

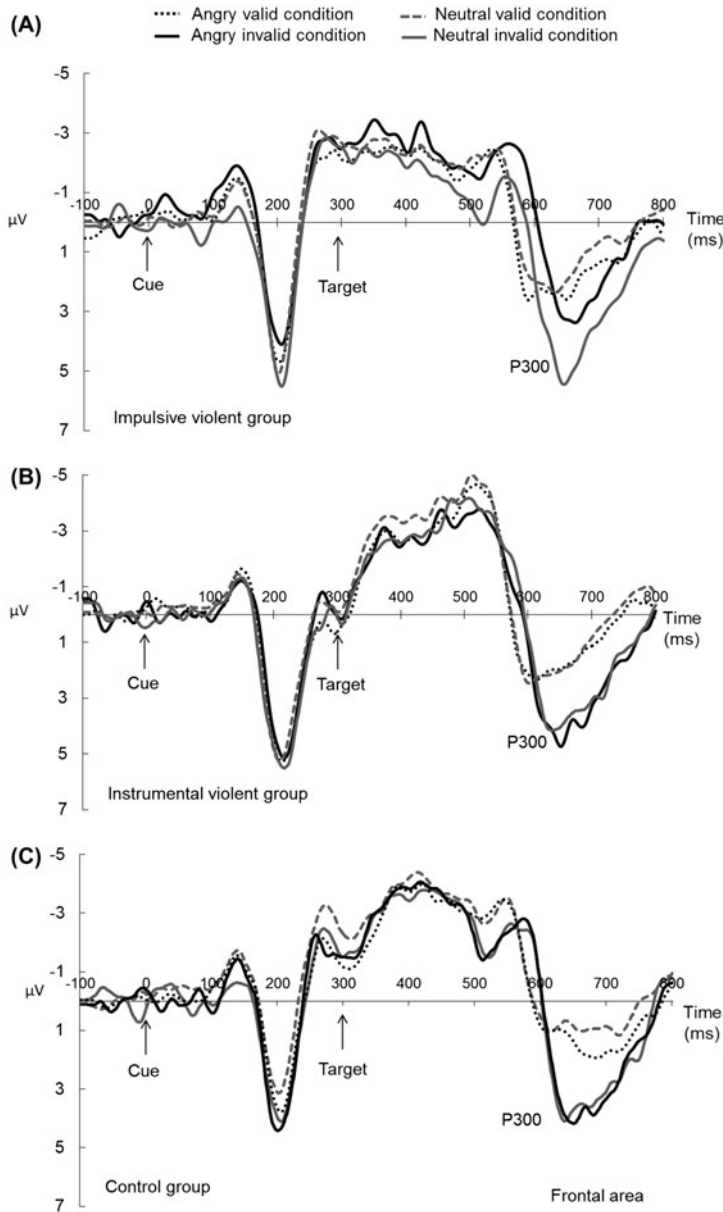


Figure 2. Grand average waveforms derived from the frontal areas for the four conditions (angry valid cues, angry invalid cues, neutral valid cues, and neutral invalid cues) in the impulsive violent group (A), the instrumental violent group (B), and the control group (C). On the angry trials, the difference between the waves for the valid and invalidly cued trials in the impulsive violent group was more negative than that of both the instrumental violent group and the control group.

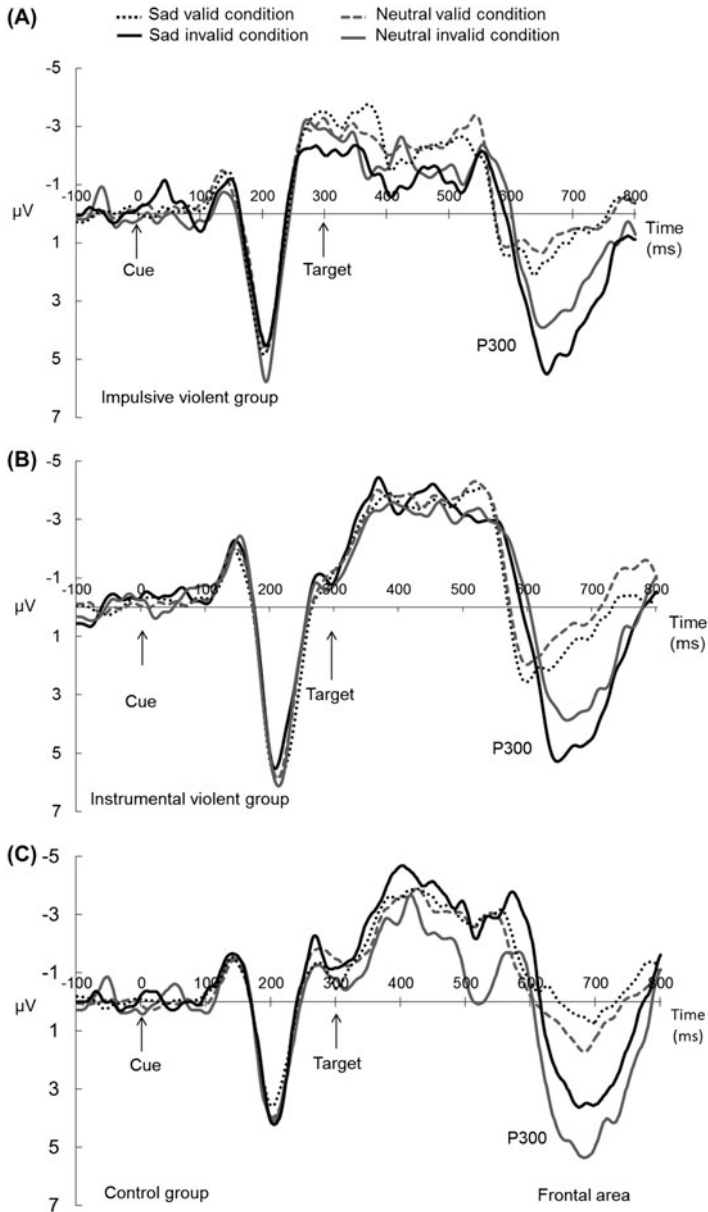


Figure 3. Grand average waveforms derived from the frontal areas by the four conditions (sad valid, sad invalid, neutral valid, and neutral invalid cues) in the impulsive violent group (A), the instrumental violent group (B), and the control group (C). For sad trials, the difference between the waves for the valid and invalid trials in the impulsive violent group and the instrumental violent group were more positive than for the matched controls.

face cue from the contralateral electrode in the angry conditions. The interval for the P300 was taken from 310 to 360 ms after target onset. To investigate P300 differences, the mean amplitude of the difference waves was computed for the ERPs from the emotional and neutral valid conditions.

No significant effects were seen on the cue evoked P100 occipital component, nor were any significant effects seen on the cue evoked N100 component. Similarly, the cue evoked P200 frontal component showed no significant effects, nor did the target-evoked P300 component on valid trials. There were no significant effects seen on the early and late N2pc difference waves. For P300 differences, the mean amplitude of the difference wave was computed for the ERPs of the emotional and neutral invalid conditions. The target-evoked P300 frontal component on invalid trials (see Figures 2–4) showed a Group $\times$ Cue valence (angry and sad) interaction effect, ($F(2,27) = 3.94$, $p = .03$). There was a significant simple main effect for group on the angry trials, ($F(2,7) = 3.51$, $p = .04$). The mean amplitude of difference waves in the impulsive violent group ($-2.06 \pm 2.09 \mu\text{V}$) showed a lesser response to angry invalid trials than that of control group ($.03 \pm 1.92 \mu\text{V}$) ($t(5(9)) = -1.9$, $p = .03$; Levene = .018, $p = .89$; Cohen's $d = 1.07$) and the instrumental violent group ($.44 \pm 2.56$) ($t(5(9)) = -2.6$, $p = .006$; Levene = 1.62, $p = .22$; Cohen's $d = 1.04$). The instrumental violent group did not differ from matched controls. On sad trials, there was a significant simple main effect for group, ($F(2,21) = 4.482$, $p = .02$). The mean amplitude of the difference waves in the impulsive violent group ($1.05 \pm 2.15 \mu\text{V}$) was more positive than that of matched controls ($-1.81 \pm 3.05 \mu\text{V}$) ($t(5(9)) = 2.3$, $p = .011$; Levene = .98, $p = .32$; Cohen's $d = 1.09$). The mean amplitude of difference waves in the instrumental

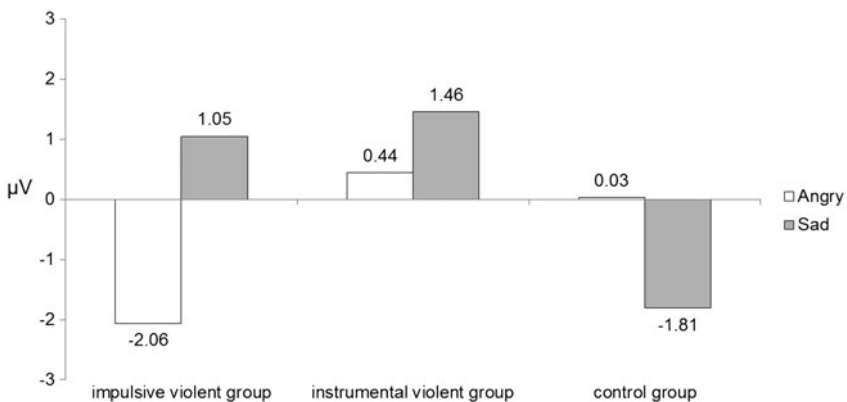


Figure 4. The mean amplitude of the P300 difference waves (in microvolts) for angry and sad conditions in the impulsive, instrumental violent, and control groups. The difference wave was computed for the ERPs for the emotional and neutral invalid conditions and the mean difference for the P300 time window calculated.

violent group ($1.46 \pm 1.69 \mu\text{V}$) was more positive than that of matched controls ($t(5(9)) = 2.9$, $p = .003$; Levene = 2.504, $p = .13$; Cohen's $d = 1.33$). There was no significant difference between the impulsive and instrumental violent groups ($t(5(9)) = -.5$, $p = .303$). It seems that impulsive and instrumental violent groups showed a greater response on sad invalid trials than the control group.

Discussion

Behavioral task

In the Posner's Cuing Task, if angry faces seize attention after they are detected in the impulsive violent group then the RT for invalid trials should be longer for these individuals than for controls. If impulsive violent offenders have a problem in disengaging attention from threatening information, then they should be slower than controls in responding to the target on invalid trials. The results showed that impulsive violent offenders did have longer RTs on invalid trials with angry faces when compared with controls. Compared with happy faces and neutral faces cues, they took longer to localize a target after an angry face cue in invalidly cued trials.

We also found that the disengagement costs in the sad face condition in the instrumental violent offenders were significantly lower than those of controls and impulsive violent offenders. Compared with happy face and neutral face cues, instrumental-violent offenders took less time to localize a target after a sad face cue in invalid trials. This potentially fits with Blair's view (Blair, 2001, 2003, 2005; Blair, Mitchell, & Blair, 2005; Mitchell, Richell, Leonard, & Blair, 2006) that instrumental violent offenders may be indifferent to sad expressions of others. The behavioral data also seem to indicate a double dissociation between impulsive violent and instrumental violent offenders in disengaging from angry and sad faces.

Behavioral and ERP data

ERP data collected during the Posner Cuing Task showed that neither impulsive nor instrumental violent offenders differed from controls in the P100 or N100 recorded from the occipital sites. These components have been associated with early attentional processing in extrastriate cortex so these findings suggest any difference in disengaging from emotional faces is not due to early differences in processing. Additionally, no effects were seen on the N2pc.

Significant differences were seen on P300 target evoked activity for the impulsive and instrumental violent offenders. In previous work (Chen et al., 2005, 2008) it was suggested that impulsive violent offenders might not have sufficient neural resources to activate the executive attentional mechanisms required to disengage from an invalid cue. When the location of the target was

opposite to the cue on the invalid trials, the anterior attention system should be activated to inhibit the engagement at the face cue and to allocate more attentional resources to disengage from the cue. The more efficiently the participants inhibit processing of the face cue, the faster the participants would disengage from the cue location. In line with this, the amplitude of the target-evoked P300 can be used to index this process. The P300 evoked by the target in impulsive violent offenders was significantly lower in the angry invalid condition, compared to the instrumental violent offenders and the controls, suggesting less attentional allocation to the target, most likely due to greater engagement by the angry face stimuli and offers a possible explanation for the behavioral performance effects. The P300 to the target in instrumental and impulsive violent offenders was higher in sad invalid trials, compared to controls, consistent with less engagement by these faces in these individuals. This P300 component, recorded from frontal electrodes, has been associated with the amount of attentional allocation generated from the ACC, so this brain region is one area that may be the source of the differences seen.

General discussion

The data obtained here illustrate clear behavioral and electrophysiological differences between instrumental and impulsive violent individuals. The instrumental offenders showed effects consistent with less interference due to any emotional stimuli. The data obtained indicated that they only showed less interference as a result of sad cues and hence seem to show no evidence of a general attentional abnormality. In contrast, impulsive violent offenders showed greater engagement by angry faces.

The reciprocal suppression model (Drevets & Raichle, 1998; Mayberg et al., 1999) offers one potential explanation for the pattern of results seen here, subject, of course, to the proviso mentioned in the introduction relating to the issues of linking ERP data to anatomical substrates. This model is based around two observations. The first is that 'emotion areas' including the amygdala, posteromedial orbital cortex, and ventral (rostral) ACC, while showing a greater response in emotion-related tasks, show decreases during some cognitively demanding tasks. The opposite is seen in other areas, including the dorsal anterior cingulate, with increased blood flow during demanding tasks, but decreases seen in both experimentally related and pathological emotional states. In the reciprocal suppression model, these two regions of the ACC are thought to interact, such that emotional information would be associated with less cognitive performance and that high cognitive demand would be associated with less processing of emotional information. According to this model, the dorsal ACC would be suppressed and rostral ACC activity would be enhanced in angry conditions. The P300, thought to originate from the ACC, in impulsive violent offenders would, in this case, be significantly lower in the angry invalid condition, when compared with the instrumental violent offenders and the

controls. The ERP data obtained from the impulsive violent offenders supported this argument. Conversely, the rostral ACC would be suppressed and the dorsal ACC would be enhanced in the sad conditions in the instrumental violent offenders. The P300 in instrumental violent offenders would be significantly higher in the sad invalid condition, when compared to the controls. Again, the ERP data from the instrumental violent offenders was consistent with this argument, although the same effect was seen for the impulsive violent offenders on this component.

Clearly, while fitting with this potential mechanism, the nature of the data obtained from frontal electrodes does not rule out other frontal areas or different mechanisms accounting for the patterns seen. Along similar lines, it would be beneficial to investigate wider emotional information and also how this interacts with emotional state in individuals like the violent offenders tested here. It is possible that some processing deficits could be manifested under different conditions (both experimentally and in terms of real-world situations).

Overall the pattern of data in the present study suggests that neither group of offenders has a general attentional deficit relating to emotional stimuli but rather that impulsive violent offenders and instrumental violent offenders may show different responses to specific types of emotional information, consistent with amygdala modulation by different classes of affective stimuli rather than by lower level information.

It is important at this point to mention some limitations of the behavioral task employed in this study, involving factors that might be valuably assessed in future work. First, the use of a cue providing no information about the subsequent target may provide a beneficial control condition for comparison of the valid and invalidly cued trials. Additionally, the use of a discriminatory response, rather than just a simple response, would allow investigation of priming effects on the data obtained.

Our data also suggest that different emotion–attention problems in impulsive and instrumental violent offenders seem to be reflected in abnormalities in disengaging attention from specific facial expressions or other emotionally significant aspects of the environment. It is important to note that further work will be necessary to both extend the breadth of the current findings and to examine their applicability to other populations and age groups.

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